

Calendar Synthesis as Poset-Feasible Padding: Dual Certificates for PSC-Q-Style Compilers

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Abstract

PSC-Q compilers choose public service calendars (slot width, renewal family, declared dither) so congestion witnesses meet a TV-tier contract. Separately, the published “poset-feasible padding” papers give dual certificates for optimal padding/scheduling under total-variation privacy constraints. This note connects the two views: PSC-Q knob selection can be phrased as a poset-feasible padding problem on a discretized time grid, so a compiler can emit an *auditable dual certificate* alongside its calendar choice. The result is practical cross-pollination: PSC-Q becomes easier to audit (a certificate explains *why* the chosen calendar is close to optimal), and the dual-certificate machinery gains a concrete deployment target (calendar compilation under load envelopes).

Scope. This is a short bridge note: it rephrases *calendar synthesis* (PSC-Q-style compilers) as an instance of *poset-feasible padding* so that deployments can publish compact *dual certificates* for their knob choices.

Imports. We import the PSC-Q mechanism and its receipt surface (Congestion series Papers 2 and 3), and the dual-certificate machinery from the poset-feasible padding line (see the 2026-01-29 wiki post).[2, 4, 5]

Exports. Exports a minimal receipt schema for calendar compilation: parameters, load envelope, witness-grid identifier, and a committed dual object whose verifier reports an optimality gap.

Artifact policy. This archive is source-only; supporting artifacts are available on request as a bundle committed by ABOM/OINL and governed by the series artifact policy (Synthesis 6).[1]

1 Motivation: “knobs” should come with a proof artifact

PSC-Q (mechanism paper) treats (δ, α) and a calendar family as compiler-visible knobs. W-Congestion-EQ (assurance paper) argues that these knobs should be receipt-visible. What remains is a compact argument that the chosen knobs are *reasonable*—not merely an arbitrary hand-tuned point. Dual certificates are the natural proof artifact: they provide lower bounds and optimality gaps without disclosing private traces.

2 A minimal discretized model

We intentionally keep the model small; full details live in the published TV-certificate papers.

Slots and service opportunities. Fix a slot width $\delta > 0$ and consider a horizon of T slots. A public calendar specifies which slot indices are *eligible* for service (and optionally a declared dither distribution for timestamps). PSC-Q’s “substitution” discipline can be idealized as: in an eligible slot, if the real queue is empty, the implementation may send cover; otherwise it serves real work.

Witness family. Let W denote a congestion witness (e.g., an RTT time series summary) extracted from a window of slots. Congestion-EQ asks for a TV-tier bound $\text{TV}(\mathcal{L}(W \mid s), \mathcal{L}(W \mid s')) \leq \varepsilon$ under a declared load envelope and auditor conditioning.

Compiler objective. A simple objective is to minimize overhead subject to meeting the witness contract. For calendar families this is often “minimize eligible-slot rate” (equivalently maximize slack) while meeting the bound for all $\rho \in [\rho_{\min}, \rho_{\max}]$.

3 Poset-feasible padding view (informal mapping)

The poset-feasible padding papers model *admissible transformations* of observable traces as a partial order (a poset) and study TV privacy under constraints. In our discretized calendar view:

- The observable trace is the slot-indexed service pattern (and any declared timestamp perturbation).
- “Feasibility” constraints are monotone: you may *delay* real work within an allowed window (scheduler constraints), and you may *fill* idle slots with cover, but you may not serve cover in a way that delays real work.
- The partial order captures allowed reschedulings and cover insertions consistent with the calendar and the substitution discipline.

Under this mapping, choosing a calendar family and parameters is choosing a feasible set in which padding/rescheduling lives.

4 Dual certificates: what the compiler can publish

The published knapsack/transport paper shows that for TV-privacy constraints over feasible transformations, one can construct dual objects that certify lower bounds and, in many cases, optimality of a candidate mechanism. We do *not* reprove those theorems here; we only state the interface that matters for PSC-Q.

Theorem 1 (Certificate interface (informal)). *Assume a PSC-Q compiler instantiates a discretized feasible set (poset constraints) and solves an optimization problem “minimize overhead subject to TV-tier constraints” on a finite witness grid. Then there exists a dual object (a set of potentials / transport dual variables) that: (i) lower-bounds the achievable overhead for the declared constraints, and (ii) yields an optimality gap for the compiler’s chosen knobs. Publishing this dual object (or a commitment to it) enables third-party auditors to check the gap without access to private traces.*

What to put in a receipt. A minimal “calendar compilation receipt” can include:

- calendar family identifier + parameters (e.g., renewal family, δ , dither fraction α),

- declared load envelope $[\rho_{\min}, \rho_{\max}]$ used for compilation,
- a witness-grid identifier (which Congestion-EQ witness summaries were constrained),
- a dual-certificate hash (commitment) and a short verifier that checks dual feasibility and reports the optimality gap.

This fits naturally into W-Congestion-EQ style receipts and transparency anchoring.

5 Practical cross-pollination opportunities

1. **From TV-certs to PSC-Q:** use dual gaps to choose between calendar families (periodic vs renewal, different dither regimes) without ad hoc tuning.
2. **From PSC-Q to TV-certs:** use “interval-robust compilation” as a standard way to define the constraint set (compile for $\rho_{\max}$ and monitor drift).
3. **From duals to monitoring:** when monitors observe regime drift, they can demand a new compilation receipt; the old dual certificate pins down how much slack was assumed.

6 Limitations

This note is a bridge: it does not claim that a simple discretization captures all congestion witnesses, and it does not replace the detailed proofs in the published poset/transport papers. The value is the *interface*: treating calendar synthesis as a certificate-emitting optimization problem.

References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] Anonymous. *PSC-Q: Public Service Calendars and Slack-Harvesting Substitution Padding for Congestion-Private Anonymous Overlays*. Mechanism paper in this unified archive.
- [3] Anonymous. *Congestion-EQ: Total-Variation Contracts for Queue/RTT Leakage in Anonymous Overlays and Anonymous DHT Lookups*. Contract paper in this unified archive.
- [4] Anonymous. *W-Congestion-EQ: Proof-Carrying Congestion Budgets with Certificates and Transparency Logs for Anonymous Overlays*. Assurance paper in this unified archive.
- [5] Anonymous. *Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy*. January 2026. Published. Wiki: [[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]].